

AI-Powered Cancer Detection: Enhancing Accuracy through Transfer Learning and Learning without Forgetting

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Introduction

The term cancer’’ describes a situation in which the body’s normal cells develop abnormally due to uncontrolled mutations. Upon creation, these cells divide erratically and disseminate throughout the organs. If left untreated, most cancers have the potential to kill us. The leading cause of death worldwide is cancer, followed by cardiovascular diseases.

Motivation

Cancer's global impact has a leading cause of death demands urgent action. Understanding cancer development and recognizing symptoms empowers individuals to take preventive measures and seek timely medical intervention.

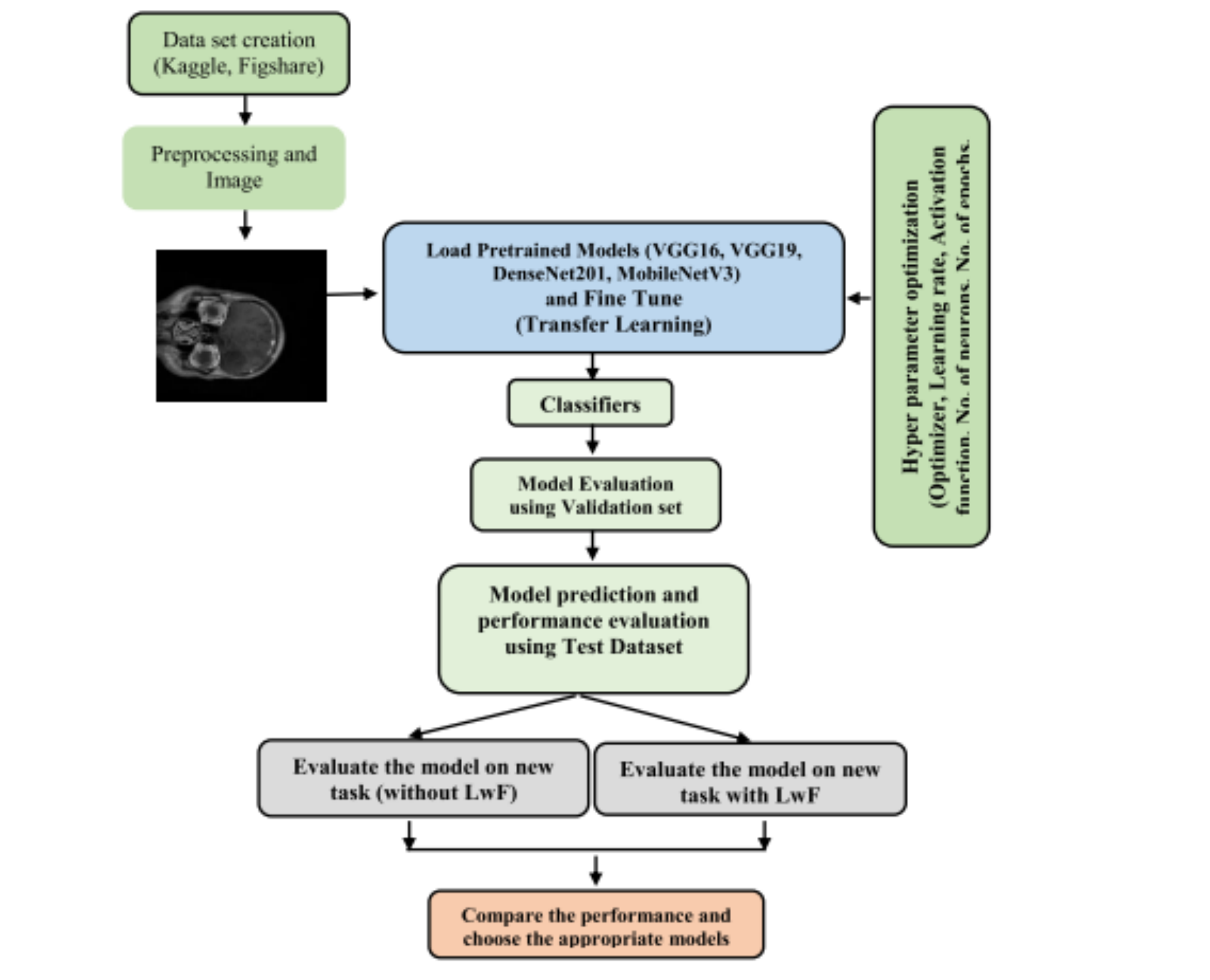
SCOPE of the Project

The purpose of understanding cancer and its diagnostic methods is to emphasize the critical need for early detection and treatment. By recognizing the factors contributing to cancer development and being aware of common symptoms, individuals can seek timely medical attention. Additionally, highlighting the significance of advanced imaging techniques like CT and MRI underscores their role in accurately detecting cancerous growths throughout the body, facilitating prompt diagnosis and intervention for better patient outcomes.

Methodology

The proposed research aims to develop AI-based deep learning models capable of accurately classifying eight types of cancer, including lung, brain, breast, and cervical cancer, utilizing CT/MRI images. The study evaluates various pre-trained Convolutional Neural Network (CNN) architectures, such as MobileNet, VGGNet, and DenseNet, for transfer learning to identify different cancer cell types. Leveraging Bayesian Optimization, the research seeks to optimize hyperparameters for enhanced model performance.

Additionally, to mitigate the risk of transfer learning leading to the forgetting of original datasets, the study employs Learning without Forgetting (LwF) techniques. LwF enables the network to retain its original capabilities while focusing on learning from new task data. By implementing LwF, the research aims to develop robust models capable of effectively distinguishing between different cancer types while ensuring that previous knowledge is not overwritten. This comprehensive approach integrates state-of-the-art deep learning techniques to advance the accuracy and reliability of cancer diagnosis using medical imaging data.



Results

The study showcases the superiority of VGG16, VGG19, DenseNet201, MobileNetV3-SMALL, and MobileNetV3-Large over existing methods, highlighting their potential for enhancing cancer classification accuracy. By implementing transfer learning and Learning without Forgetting (LwF) techniques, the models exhibit improved adaptability and mitigate knowledge transfer issues, contributing to enhanced performance in cancer detection.

Moreover, the extension of the Xception model significantly enhances prediction accuracy, further bolstering the capabilities of the proposed approach.

Model	Accuracy	Recall	Precision	F1
VGG16	0.995106	0.994683	0.995438	0.995056
VGG19	0.952413	0.950212	0.955314	0.952731
DenseNet201	0.988096	0.987721	0.988424	0.988069
MobileNetV3-Small	0.995827	0.995769	0.995942	0.995855
MobileNetV3-Large	0.998567	0.998519	0.998596	0.998557
Xception	0.990913	0.990596	0.991185	0.990887

Conclusion

The proposed AI-based deep learning models offer a promising solution for automating cancer detection tasks, with the potential to significantly improve patient outcomes through early and accurate diagnosis. By leveraging transfer learning, optimized hyperparameters, and LwF techniques, these models demonstrate superior performance in classifying various cancer types from CT/MRI images. The integration of these models into clinical practice, facilitated by userfriendly interfaces, can empower healthcare professionals and promote equitable access to advanced cancer detection technologies. Continued research and collaboration with stakeholders in the medical community are recommended to further refine and expand the applicability of these models, ultimately contributing to improved cancer care and patient wellbeing.

References

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